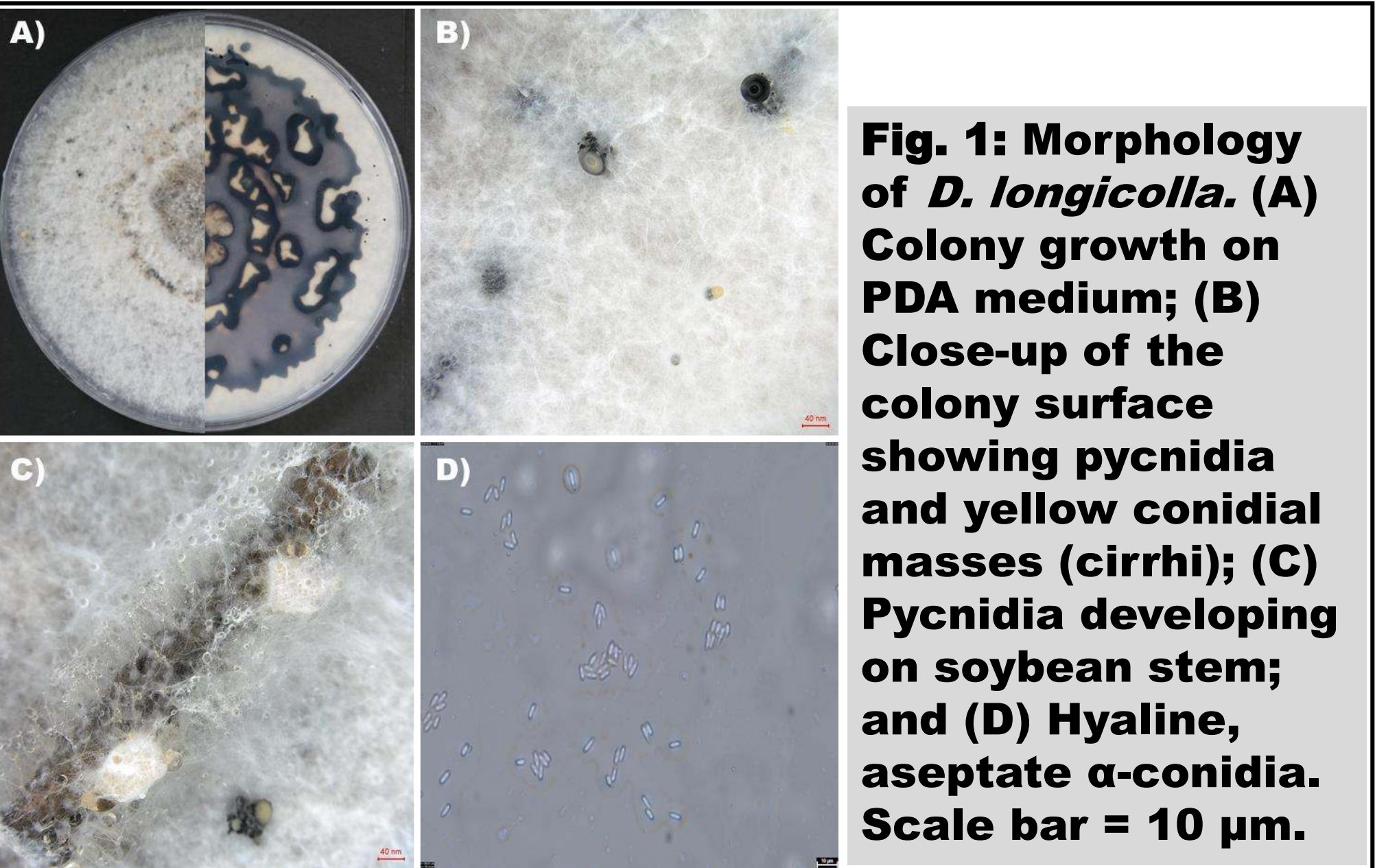


Abstract

We aimed to develop a seedling-based pathogenicity assessment method that enables efficient screening of large soybean varieties under greenhouse or growth-chamber conditions, overcoming the lengthy assessment period required by conventional end-of-season evaluation of pod and stem blight caused by *Diaporthe longicolla*. First, eight inoculation methods were evaluated, including mycelial plug inoculation, toothpick inoculation, spore injection, conidial suspension injection, mycelial spraying, stem-wound inoculation, mycelial contact inoculation, and ascospore inoculation. The methods were compared using disease severity index (DSI%) and area under the disease progress curve (AUDPC), and the mycelial plug inoculation method was identified as the most effective approach for large-scale screening of soybean varieties.

Materials and Methods

Colony morphology of *D. longicolla*



Eight seedling-based inoculation methods

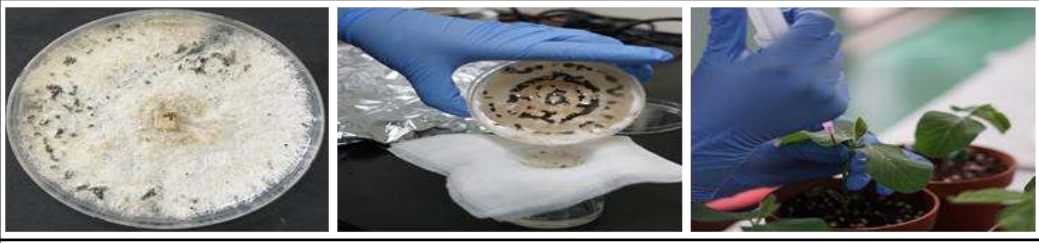
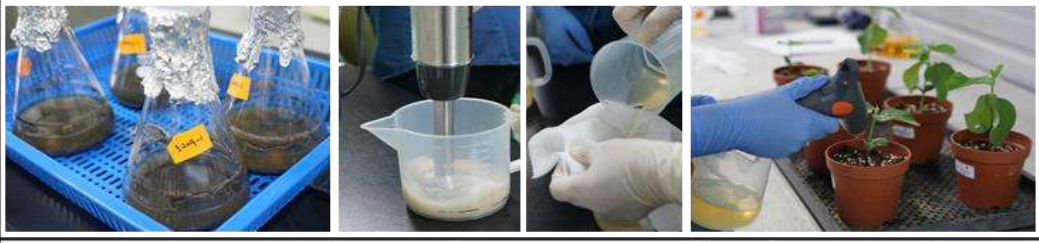
Inoculation methods	Inoculation procedures
Toothpick inoculation	
Stem-wound inoculation	
Mycelium contact inoculation	
Cut-seedling inoculation	
Spore injection	
Conidial suspension injection	
Spraying of mycelium	
Ascospore inoculation	

Fig. 2: Seedling-based inoculation methods used in this study.

Conclusions

- ◆ The cut-seedling inoculation method provides a rapid, efficient, and reliable approach for screening soybean varieties.
- ◆ DS% and AUDPC varied significantly among the inoculation techniques.

Acknowledgements

This research was supported by a grant from the Agenda project of the Rural Development Administration (RDA) (project number: PJ01740702), Republic of Korea

Results

Stem lesion length of soybean seedlings inoculated with *D. longicolla* using 8 inoculation methods

Inoculation methods	Mean stem lesion length (cm)			
	GJ5209		YC7743	
	Taekwang	Pyeongwon	Taekwang	Pyeongwon
Toothpick inoculation	3 ^{cd}	2.2 ^{de}	2.5 ^d	2.2 ^{de}
Stem wound inoculation	3.2 ^{cd}	3.2 ^{cd}	2.5 ^d	2.5 ^d
Mycelial contact inoculation	2.3 ^{de}	1.2 ^f	0.9 ^f	0.3 ^f
Cut-seedling inoculation	5.1 ^a	4.3 ^b	3.8 ^{bc}	3.8 ^{bc}
Spore injection	1 ^f	1.3 ^{ef}	1.4 ^{ef}	1 ^f
Conidial suspension injection	1.6 ^{ef}	1.5 ^{ef}	1.6 ^{ef}	2 ^{de}
Mycelial spray inoculation	0	0	0	0
Ascospore inoculation	0	0	0	0

Honestly Significant Difference (HSD) = 1.65

Symptoms of pod and stem blight

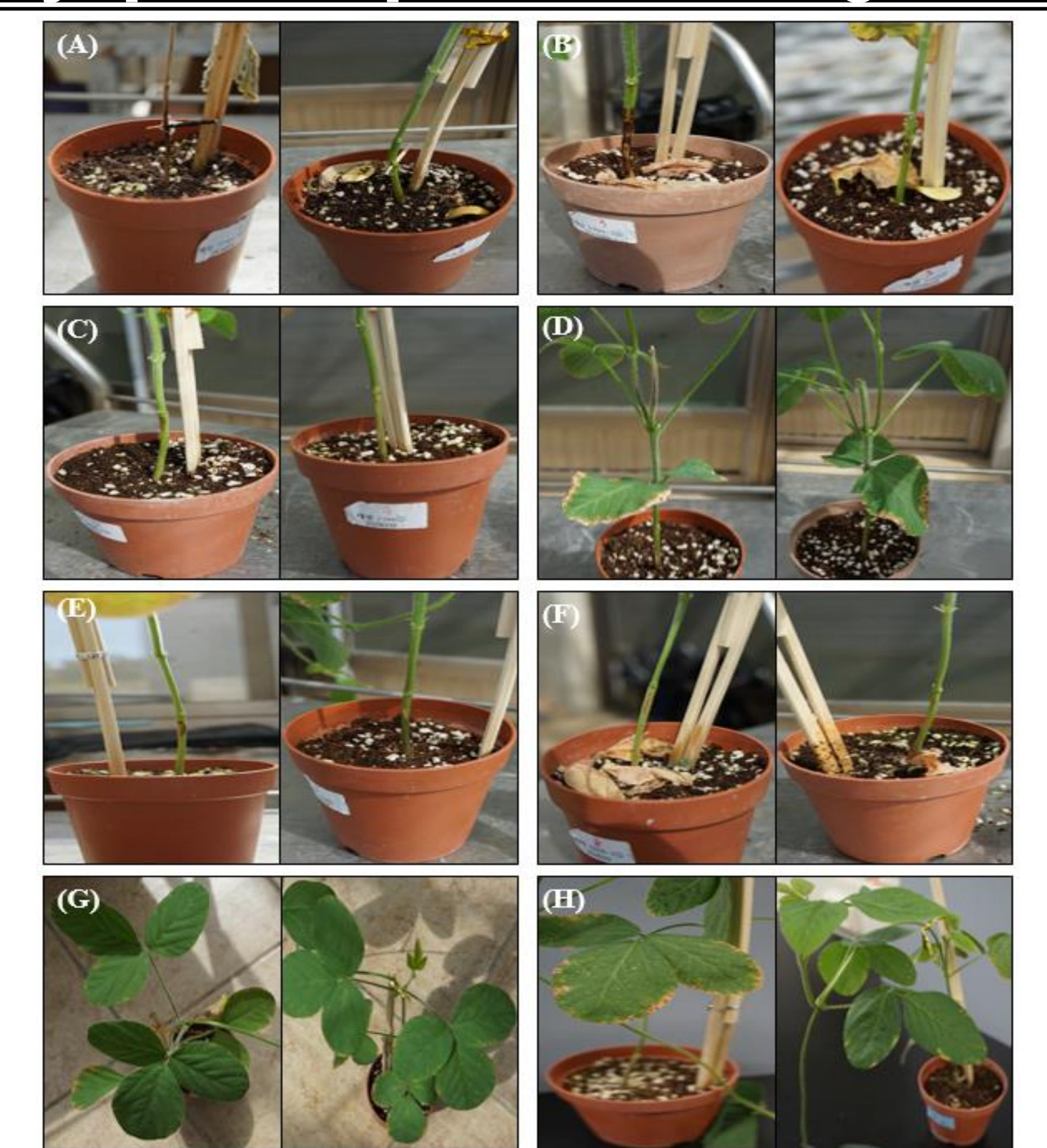


Fig. 3: Soybean seedlings inoculated with: (A) Toothpick, (B) Stem-wound, (C) Mycelium contact, (D) Cut-seedling, (E) Spore injection, (F) Conidial suspension injection, (G) Spraying of mycelium, and (H) Ascospore inoculation methods at 14 DAI.

References

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Li, S. 2018. Development of a seedling inoculation technique for rapid evaluation of soybean for resistance to *Phomopsis longicolla* under controlled conditions. *Plant Methods*, 14, 81-101.

Disease severity of *D. longicolla* in soybean

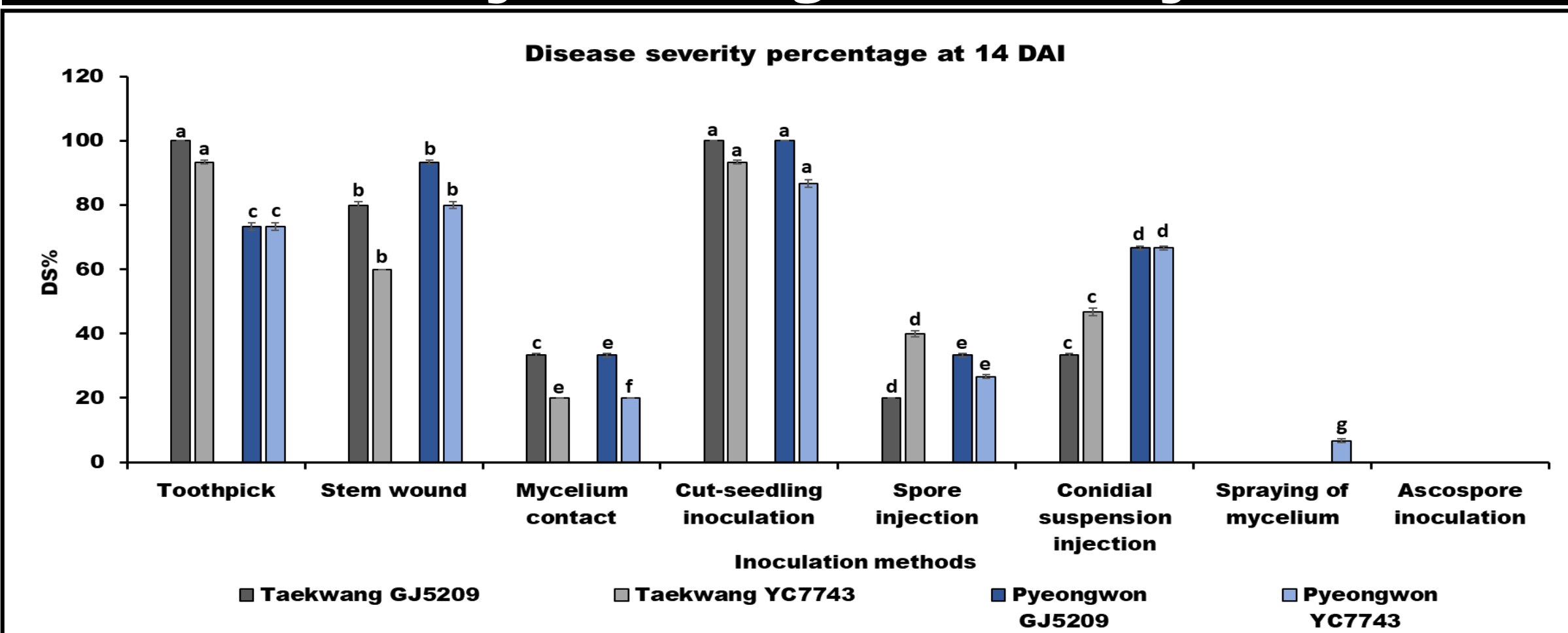


Fig. 4: Disease severity percentage in soybean seedlings inoculated with *D. longicolla* isolates using various methods, calculated at 14 DAI. Significant at $p < 0.0001$.

AUDPC of *D. longicolla*, 4-14 DAI progress

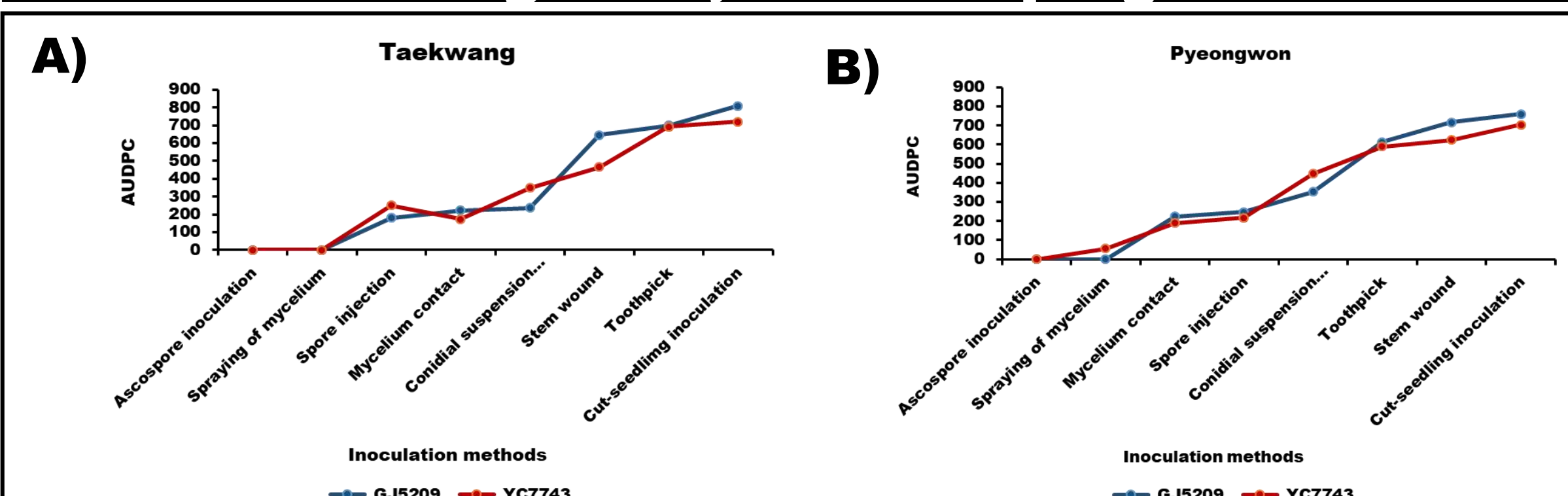


Fig 5: AUDPC of pod and stem blight disease on (A) Taekwang and (B) Pyeongwon varieties. Significant at $p < 0.0001$